

⚡ AI-POWERED CYCLONE RESILIENCE PLATFORM

PRAHAR

Predictive Risk & Anticipatory Hazard Action Resource

Track-Based Cyclone Impact & Infrastructure Vulnerability Forecaster for Bay of Bengal & Coastal APAC

🌐 Google Earth Engine 30m

⚡ Gemini 1.5 Flash Multimodal

🌊 Hydrodynamic Bathymetry

🇮🇳 22 Indic Languages Audio

Designed for District Magistrates, NDRF Commanders & State Disaster Authorities (OSDMA, WBDMD, APSDMA, TNDMA) to trigger anticipatory evacuations < T-12h before landfall.

Core Objectives



1. Ingest Track Telemetry

Real-time ingestion of IMD bulletins, JTWC advisories, tide gauges, and Copernicus 30m GEE DEM elevation models.



2. Surge Modeling

Hydrodynamic coastal inundation envelopes across 3 surge hazard tiers (>2.5m, 1.5–2.5m, 0.5–1.5m).



3. Infrastructure Graph

Pinpoint severed CHCs, tripped substations, inundated roads, and cutoff cyclone shelters before surge strikes.



4. Multimodal SOPs

Gemini 1.5 Flash auto-drafts actionable tactical directives for NDRF, DMs, and Panchayats in 22 Indic tongues.



5. Parametric Escrow

Automated pre-landfall liquidity escrow releasing emergency disaster cash at T-24h with zero claims latency.



Our Mission: Empower State & District Disaster Authorities with track-based predictive intelligence, multimodal AI, and transparent parametric liquidity — turning raw forecast data into life-saving anticipatory action.

PROBLEM STATEMENT

The Coastal Vulnerability Gap

● The Problem Today

▲ Fragmented Telemetry Silos

Cyclone tracks from IMD, bathymetry from INCOIS, power grids, and hospital directories exist in isolated databases without unified spatial correlation.

▲ 12–24h Bureaucratic Response Lag

Standard disaster dashboards update post-facto. Ground damage assessments occur 24–48 hours after coastal embankments are breached.

▲ Static Paper–Based Emergency SOPs

District Collectors rely on generic paper manuals unsuited for live dynamic water levels, leading to trapped civilians and flooded evacuation routes.



"The storm surge inundated the coastal arterial highway before the evacuation buses were dispatched."

Without predictive, track-based infrastructure vulnerability modeling, relief forces deploy blindly after lifelines sever. Coastal districts lose critical power and emergency medical access within hours.

The PRAHAR Solution

1. Unified Geoprobe

Direct ingestion of live IMD track coordinates, JTWC advisories, Copernicus 30m DEM elevation, and OpenStreetMap infrastructure networks into a unified spatial graph.

2. Multimodal AI Triage

Gemini 1.5 Flash evaluates hydrological choke points, drafting role-directed SOPs and generating multi-lingual Web Speech broadcasts across 22 official Indian languages.

3. Parametric Liquidity

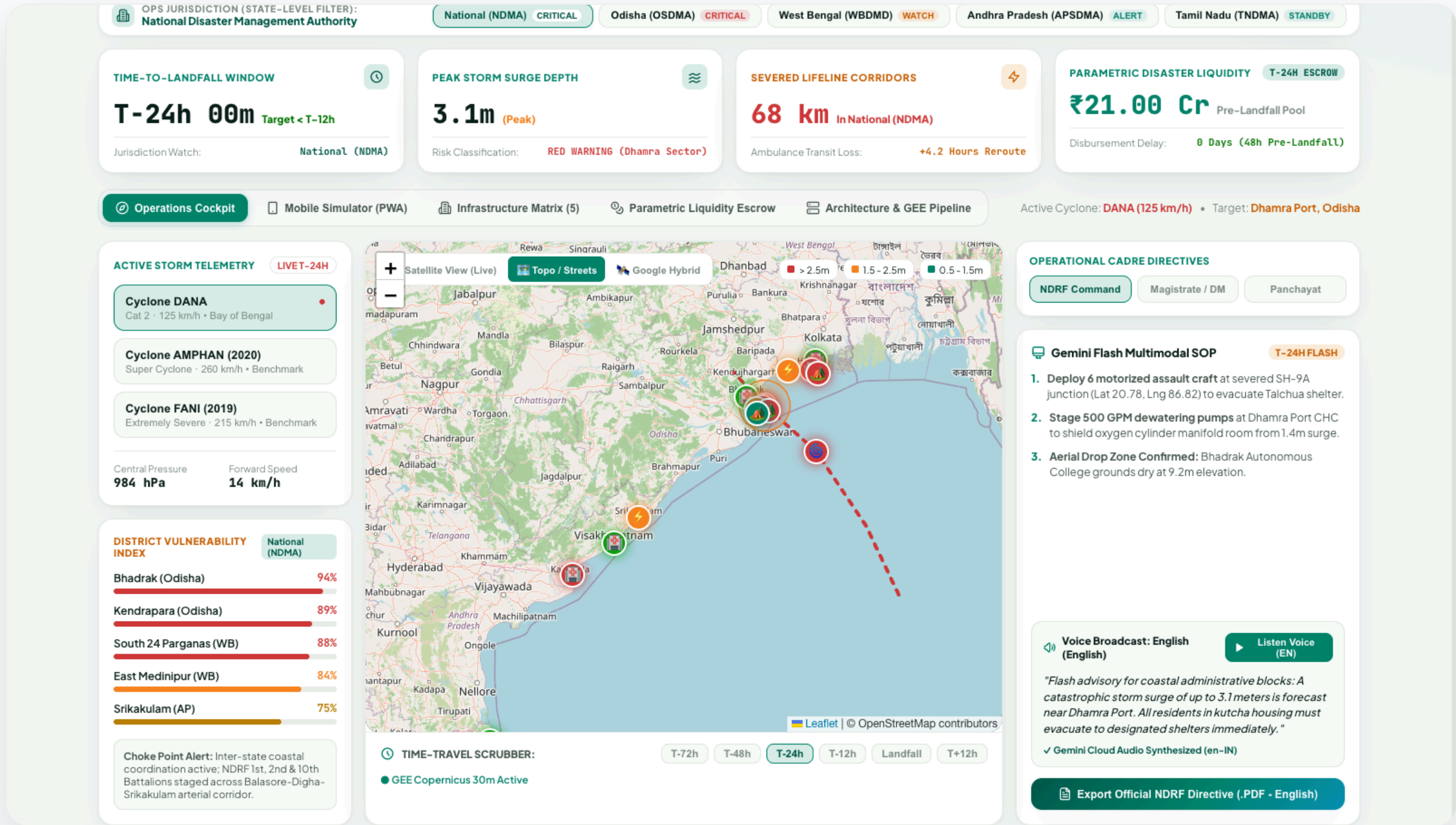
Automated pre-landfall escrow releases emergency cash reserves at T-24h directly to SDRF accounts without bureaucratic verification delays or post-facto claims.



The 5–Minute District Collector Value: You don't have to decipher raw radar files. PRAHAR instantly highlights which bridge is severed, which hospital loses backup power, and where amphibious assault craft must be pre-staged.

INTERACTIVE SHOWCASE

Live Operations Cockpit



Real-Time Capabilities

• Dynamic Storm Tracking

Live trajectory of Cyclone DANA (125 km/h) with 6-stage temporal scrubber from T-72h to Landfall.

• 3-Tier Surge Envelopes

Catastrophic (>2.5m red), Severe (1.5–2.5m orange), and Saline (0.5–1.5m green) inundation polygons.

• District Vulnerability Index

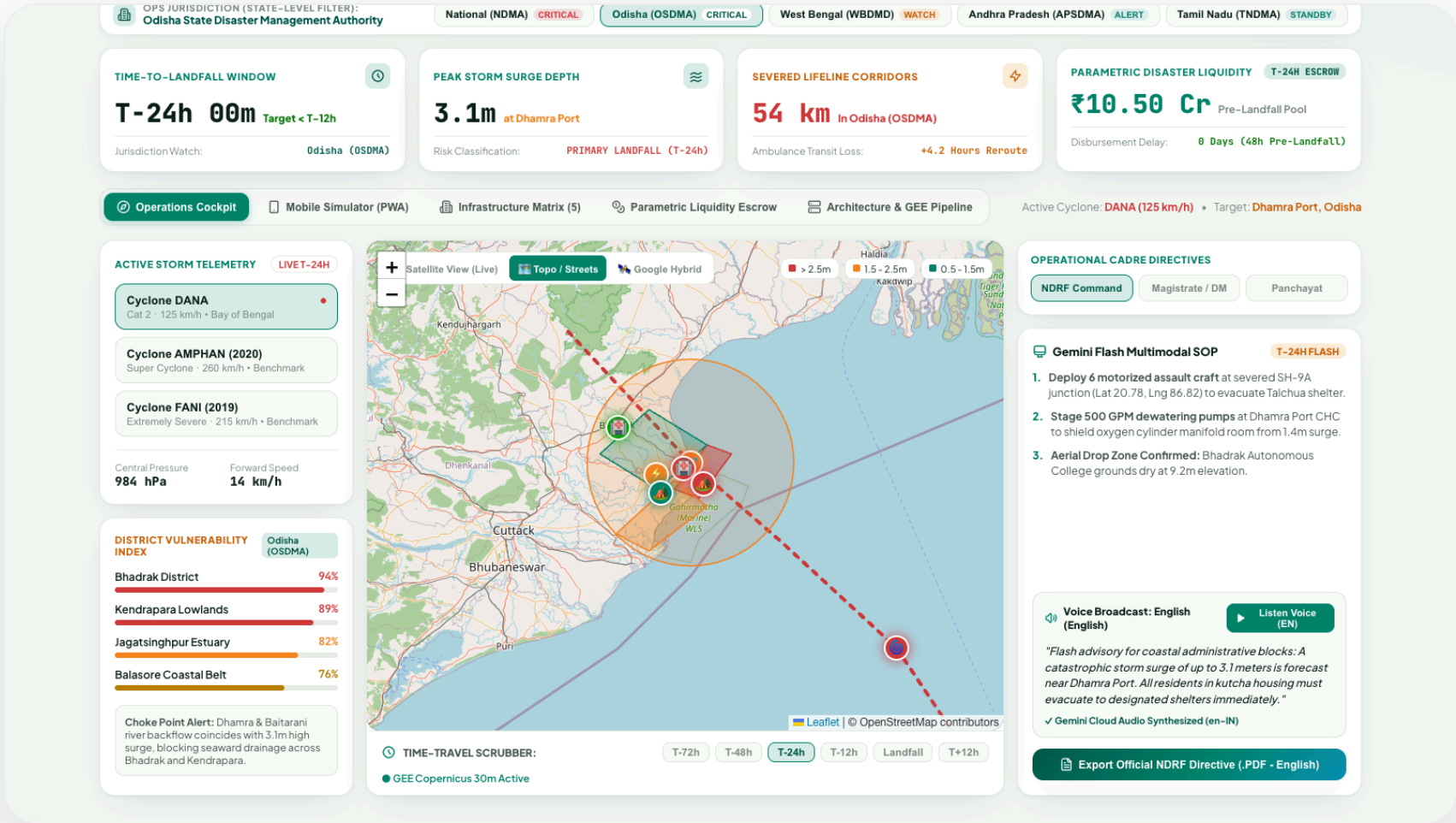
Ranks Bhadrak (94%), Kendrapara (89%), and South 24 Parganas with hydrological choke point alerts.

• Multimodal SOP Directives

Role-based tactical instructions for NDRF, District Collectors, and Panchayats.

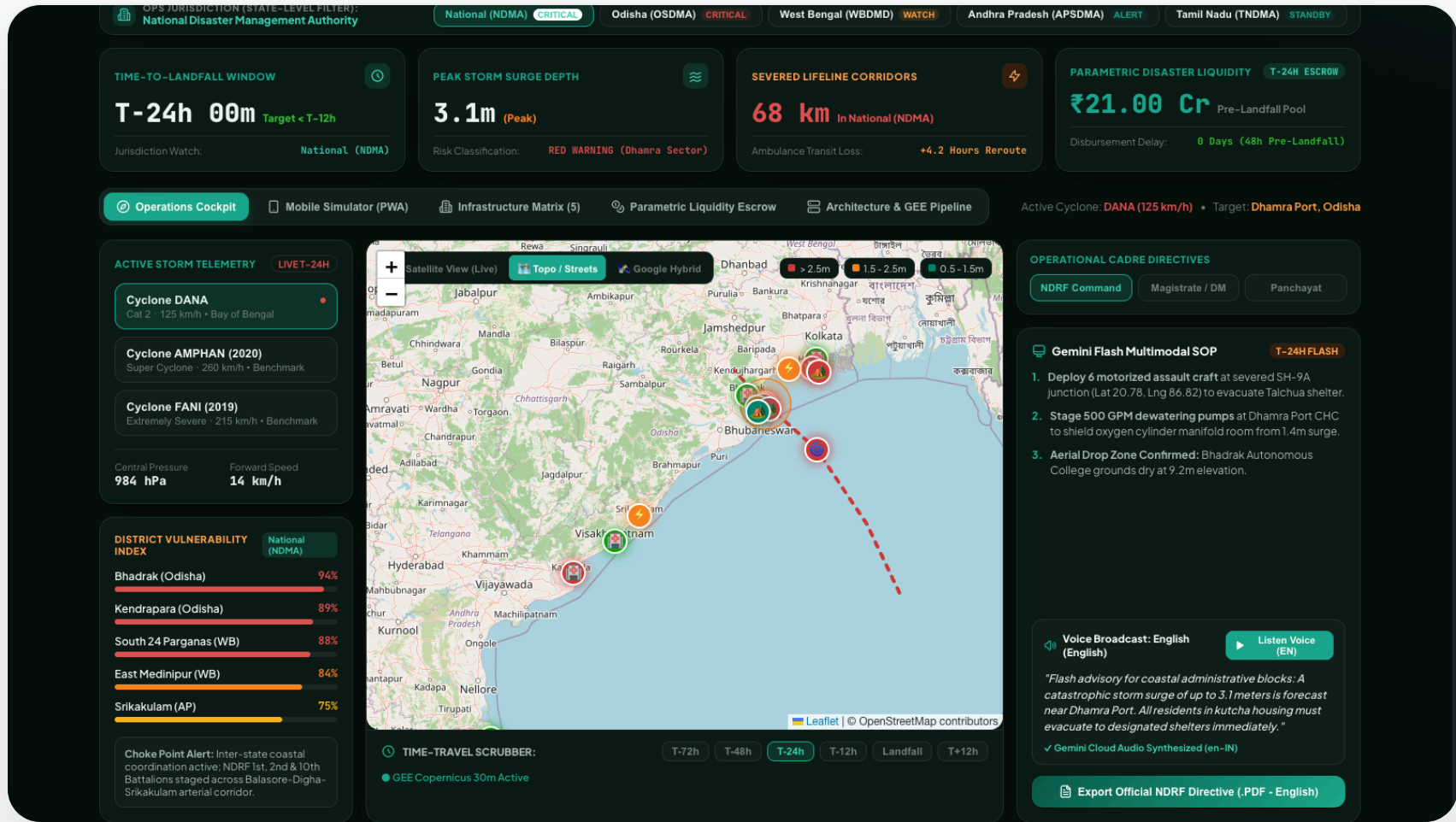
Integrated with Leaflet GIS & OpenStreetMap Topo View

State Operations & 22 Official Indic Languages



Multi-State Operational Jurisdictions

1-click filtering across National (NDMA), Odisha (OSDMA), West Bengal (WBDMD), Andhra Pradesh (APSDMA), and Tamil Nadu (TNDMA) automatically re-centers GIS and updates state-specific choke points and asset markers.

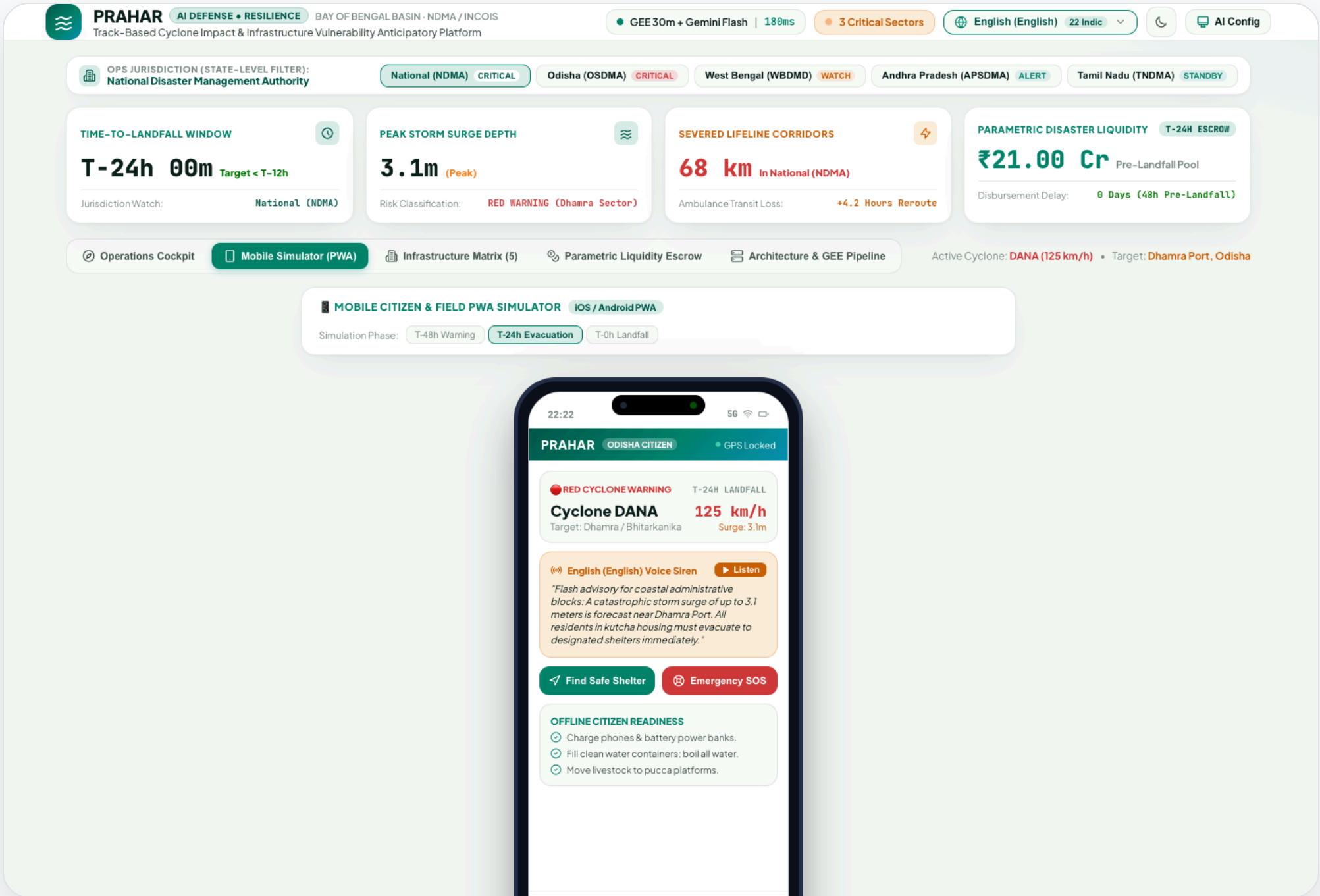


All 22 Eighth Schedule Indian Languages

Native translations and instant Web Speech audio broadcasts for Odia, Bengali, Telugu, Tamil, Hindi, Marathi, Gujarati, etc. Dark Mode supports 24/7 command center screening without visual fatigue.

CITIZEN LAST-MILE

Mobile Citizen Simulator (PWA)



📱 Citizen Protection Features

✓ Offline-First PWA Architecture

Pre-caches elevation maps and shelter waypoints. Fully functional even when mobile towers lose power or backhaul connections.

✓ Dynamic Shelter Navigation

Real-time routing diverts citizens away from flooded roads toward elevated shelters with verified remaining capacity.

✓ One-Tap SOS Distress Beacon

Broadcasts GPS coordinates, battery level, and family count to district emergency operations centers.

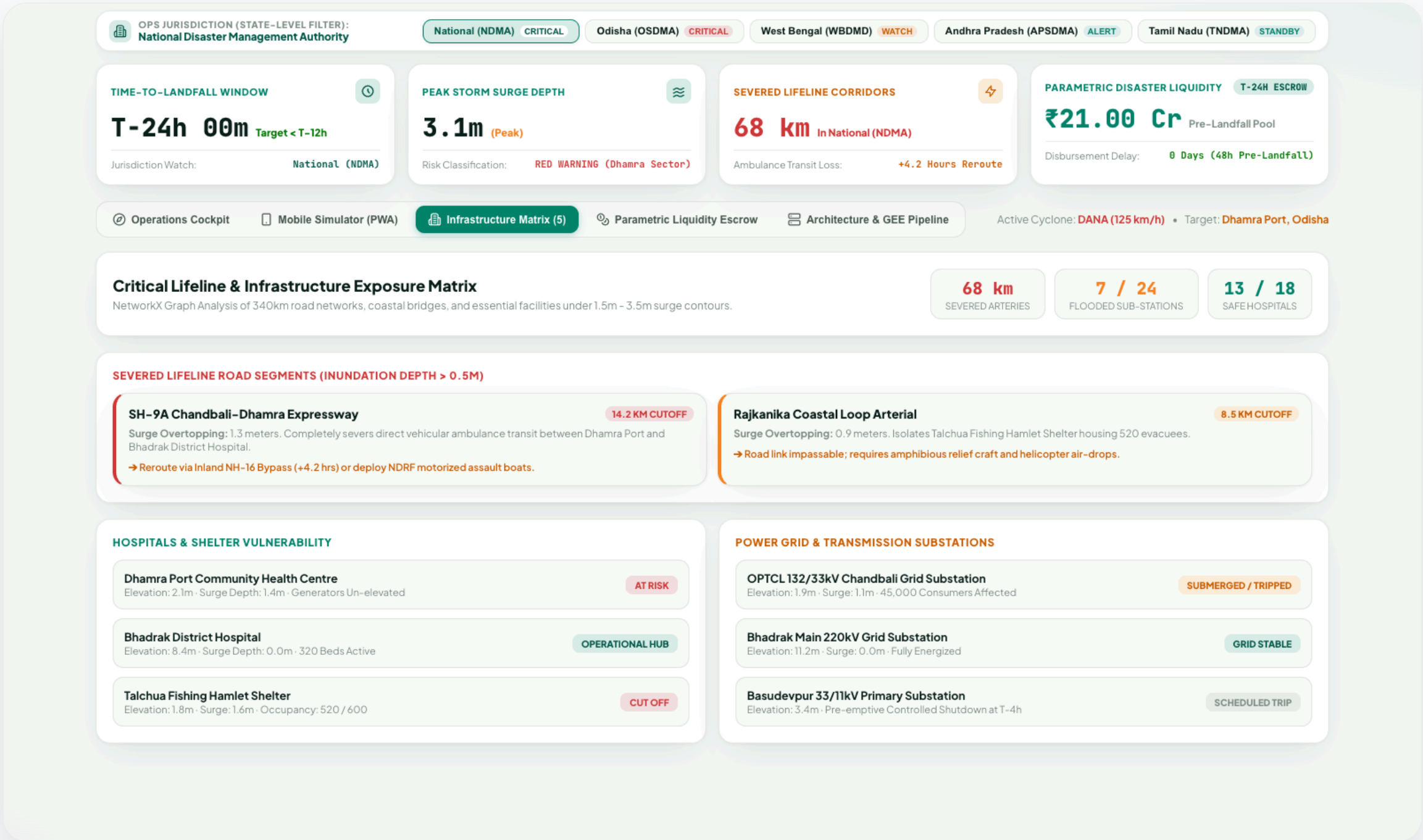
✓ Parametric Relief QR Pass

Digital QR credentials enable immediate ration and relief fund verification at local evacuation shelters.

Simulated on iPhone 16 Chassis with Dynamic Island Telemetry

EXPOSURE ASSESSMENT

Infrastructure Vulnerability Matrix





Automated Vulnerability Index

$$V_dist = 0.45 \cdot (S_peak / E_mean) + 0.35 \cdot (C_sev / C_tot) + 0.20 \cdot p_kutcha$$

● Zero Claims Processing

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● Pre-Landfall Disaster Liquidity

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● Cryptographic Auditability

● Cryptographic Auditability

48h Pre-Landfall Liquidity Disbursed to State Accounts

Tab 4: Parametric Liquidity Escrow (?tab=parametric)

Enterprise System Architecture & Immediate Readiness

PRAHAR SYSTEM ARCHITECTURE & DATA PIPELINES

Built with an enterprise decoupled microservice architecture: real-time telemetry ingestion operates on dedicated asynchronous pollers, while the high-performance FastAPI core engine powers hydrodynamic flood models and Gemini Flash multimodal spatial reasoning.



1. Ingestion Microservice

Port 8001 · Async Poller

- **IMD & JTWC ATCF:** Real-time cyclone coordinates, central pressure, and radii.
- **OpenStreetMap Overpass:** Live vector extraction of hospitals, power grids, and highways.
- **INCOIS Telemetry:** Tidal station tables and sea surface temperature anomalies.



2. Google Earth Engine (GEE)

Satellite Raster Pipeline

- **Copernicus GLO-30 DEM:** 30-meter precision coastal elevation gradients.
- **Sentinel-1 SAR:** Synthetic aperture radar backscatter for past flood calibration.
- **GPM IMERG:** Multi-satellite precipitation accumulation for compound runoff.



3. Simulation & Graph AI

FastAPI · Port 8010

- **SLOSH Hydrodynamic Model:** Inverted barometer setup & shallow coastal bathymetry slope.
- **Compound River Backflow:** Intersects ocean surge with inland delta drainage channels.
- **NetworkX Graph:** Dynamically severs roads submerged >0.5m; computes shelter isolation index.



4. Gemini Flash Reasoning

Zero-Latency Spatial SOPs

- **Multimodal GIS Ingestion:** Evaluates satellite composite images & contour rasters.
- **Role-Based SOP Synthesis:** Targeted briefs for NDRF, District Magistrates, and Citizens.
- **Parametric Smart Escrow:** Triggers 48h pre-landfall liquidity without bureaucracy.

REAL-TIME LATENCY & ANTICIPATORY ACTION BENCHMARKS

GEE Elevation Fetch

1.2 sec

Hydrodynamic Surge Sim

840 ms

NetworkX Graph Severance

320 ms

Gemini Flash Reasoning

620 ms

Anticipate. Protect. Empower Coastal Resilience.

Deployable across National, State & District Disaster Authorities in under 48 hours with zero structural changes to legacy data.

⚡ Ingestion Poller (Port 8001) • ⚡ FastAPI Core Engine (Port 8010) • ⚡ Vite GIS Command Center (Port 5174) • ⚡ 100% Free Open Baseline